



Evaluation of Cellular Connectivity Under Mobility

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Agenda

Project Overview

Motivation, Objectives, Problem Definition

Simulation Setup

MATLAB Framework, Modular Structure, Cellular Deployment Modes, Mobility Scenarios & Speed Profiles, Modeling Doppler Effect & Monte Carlo Averaging, Base Station Deployment Modes

Evaluation Metrics

SINR, Throughput, Handover Performance, Radio Link Failure

Results & Analysis

Conclusions

Project Overview

Motivation

Cellular links often degrade when users move at high speeds.

Different base-station deployment modes may behave very differently under mobility.

Objectives

Evaluate communication quality under various speeds.

Compare performance across multiple base-station configurations.

Identify which mode offers the most stable connectivity under mobility.



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Project Overview

Problem Definition

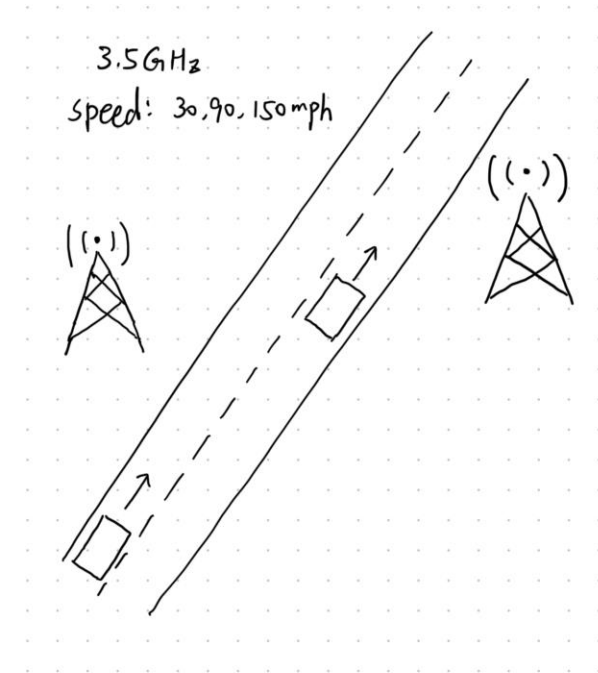
Carrier Frequency = 3.5 GHz

The user equipment moves along a straight trajectory at constant speed.

Three representative mobility levels are considered: 30, 90, and 150 mph.

Simulations are performed with and without MIMO

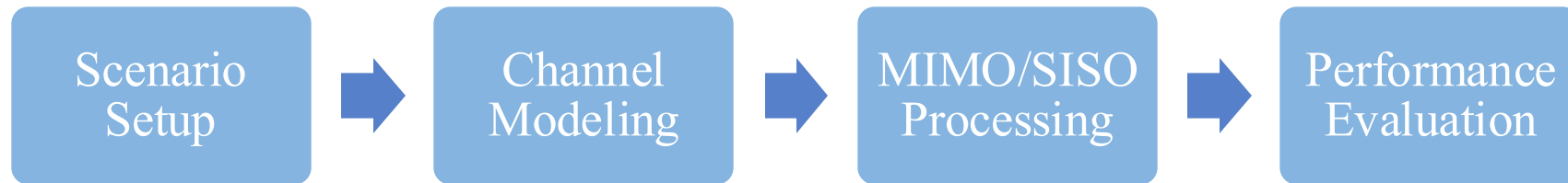
Communication quality is evaluated under these controlled conditions across different base-station deployment modes.



Simulation Setup

MATLAB Framework

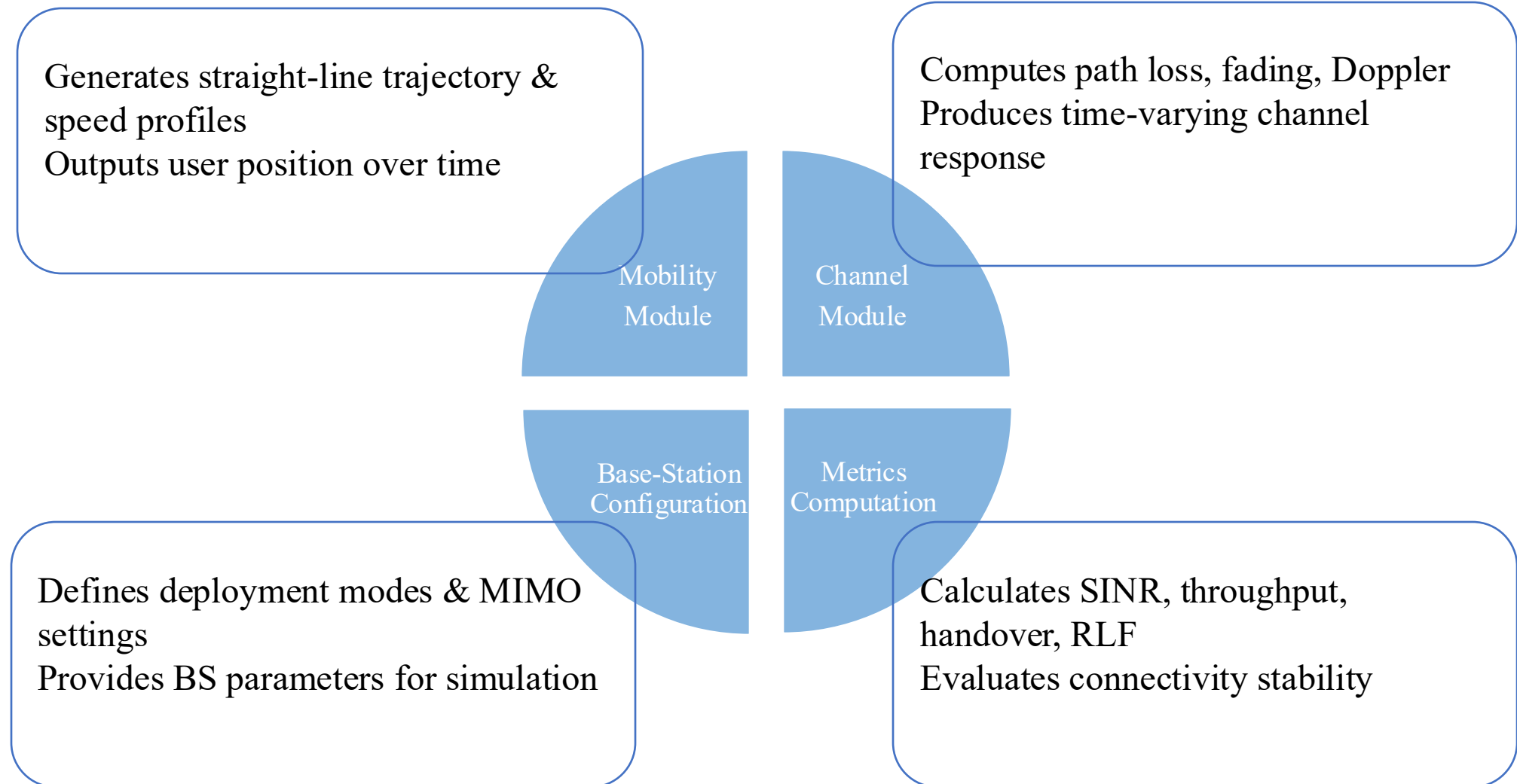
Simulation Pipeline



MATLAB Framework

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Modular Structure



MATLAB Framework

Configurable parameters

Speeds (30/90/150 mph)

deployment modes

MIMO on/off

Outputs

SNR

Throughput

Handover

RLF

Cellular Deployment Modes

Macro Cell Deployment

Large coverage, high transmit power

Good for wide-area mobility (e.g., highways)

Fewer handovers but signal fades quickly when far from the tower

Strongly affected by speed in terms of Doppler + channel variation

Small Cell Deployment

Low-power, dense layering (pico / femto cells)

Higher capacity, especially in hotspot areas

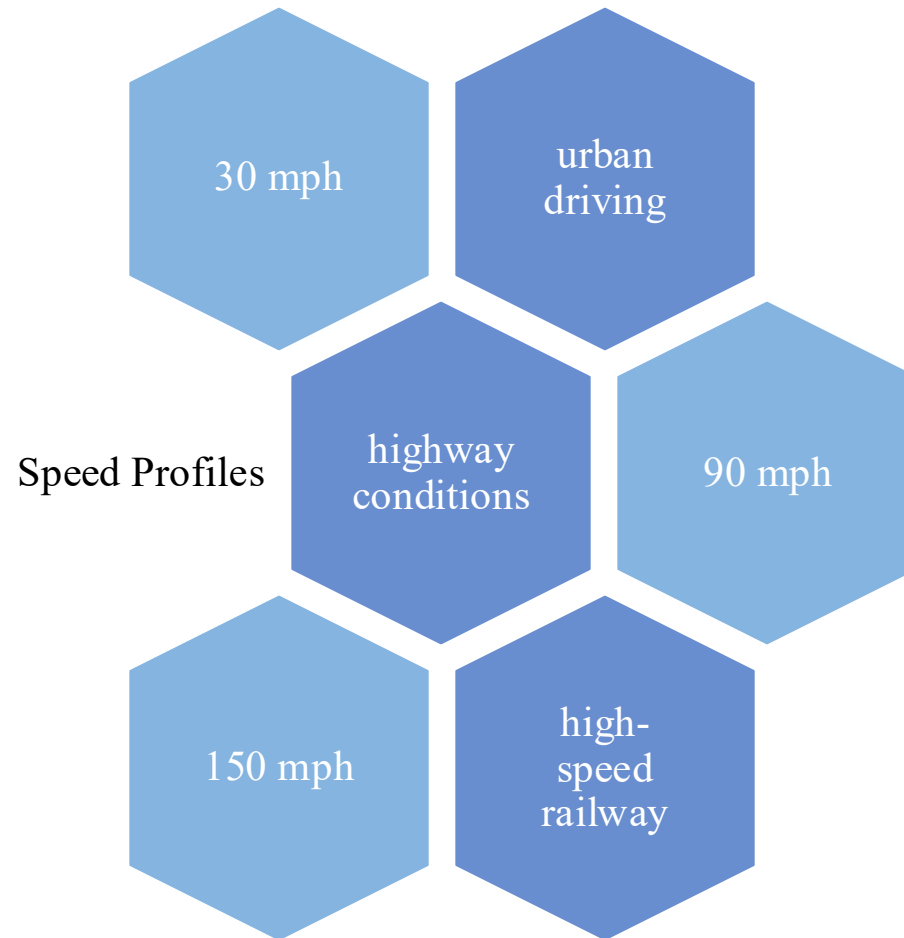
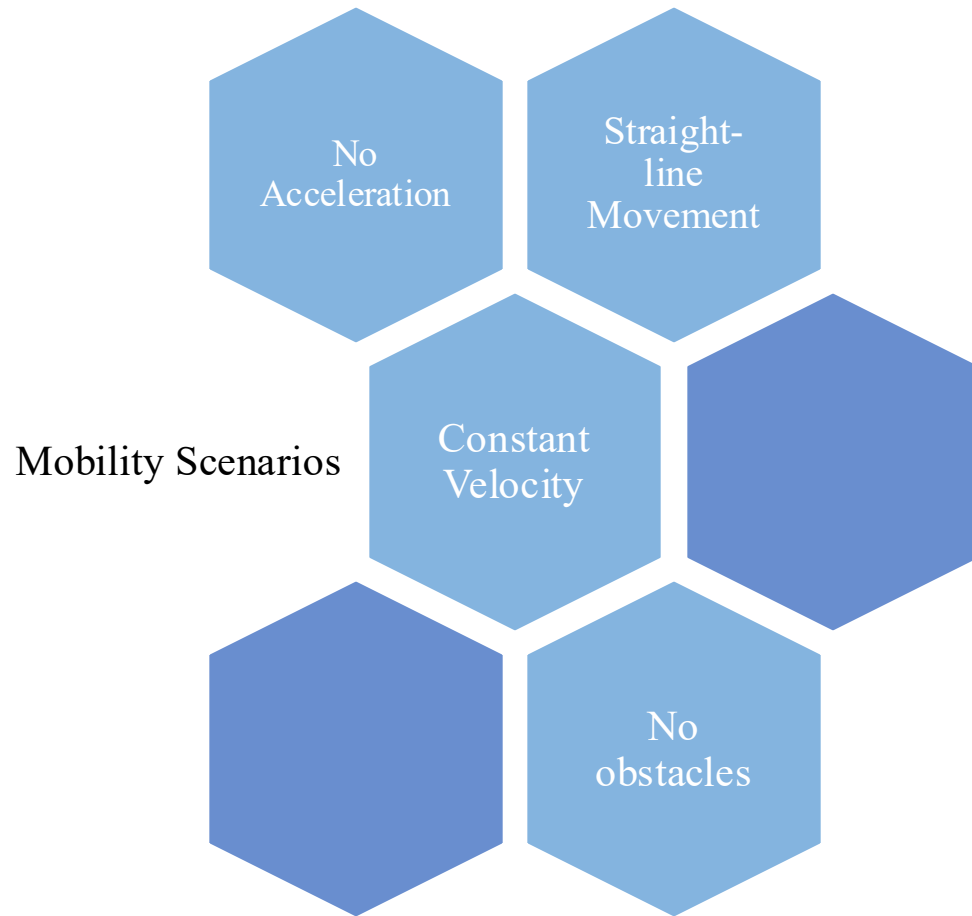
But mobility challenges:

Frequent handovers

Increased risk of Radio Link Failure (RLF)

Typically less suitable for high-speed scenarios unless optimized

Mobility Scenarios & Speed Profiles



Modeling Doppler Effect & Monte Carlo Averaging

Doppler Effect Simulation

UE mobility causes time-varying small-scale fading.

Maximum Doppler frequency:

$$f_D = \frac{v}{\lambda} \quad \lambda = \frac{c}{f_c}$$

Time-correlated Rayleigh fading generated using:

Complex Gaussian sequences per BS.

Low-pass filter with cutoff $\sim 1.2 \times f_D$ to simulate Doppler correlation.

Output: fadingCorr matrix (BS $\times$ time), incorporated in SINR & throughput calculations.

Modeling Doppler Effect & Monte Carlo Averaging

Monte Carlo Averaging

Small-scale fading and shadowing are random.

Multiple realizations ($N_{mc} = 10$) to reduce stochastic fluctuations.

Metrics (SINR, throughput, HO/RLF counts) averaged over all realizations:

$$Metric_{avg} = \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} Metric_i$$

Base Station Deployment Modes

Macro-only

Number of BSs: 3 macro BSs
Positions: -1000, 0, 1000
Transmit Power: 46
dBm(~40W)
Shadow Fading std: 12 dB

Macro + Small

Number of BSs:
3 macro BSs + 2 small cells
Positions: -1000, -500(small
cell), 0, 500(small cell), 1000
Transmit Power:
46 dBm(~40W), 30 dBm(~1W)
Shadow Fading std:
12 dB(macro), 8dB(small)

Dense-Small

Number of BSs: 11 small cells
Positions:
every 200 from -1000 to +1000
Transmit Power: 30 dBm(~1W)
Shadow Fading std: 8 dB

Evaluation Metrics

SINR (Signal-to-Interference-plus-Noise Ratio)

Definition: Measures the quality of the received signal relative to interference and noise. Higher SINR $\rightarrow$ better link quality.

Code Measurement:

```
P_signal_lin = 10^(servPow/10);  
P_int_lin = sum(10.^(Rx_dBm/10)) - P_signal_lin;  
P_noise_lin = 10^(noise_power_dBm/10);  
SINR_lin = P_signal_lin / (P_int_lin + P_noise_lin);  
SINR_dB = 10*log10(SINR_lin);|
```

Evaluation Metrics

Throughput

Definition: Maximum achievable data rate (bps) at the UE given current SINR.

Code Measurement:

```
throughput_Mbps = BW * log2(1 + SINR_lin) / 1e6;
```

Handover Performance (HO)

Definition: Number of times the UE switches from one serving BS to another.

Code Measurement:

```
if tt > 1 && serving_idx(tt) ~= serving_idx(tt-1)
    ho_count = ho_count + 1;
end
```

Evaluation Metrics

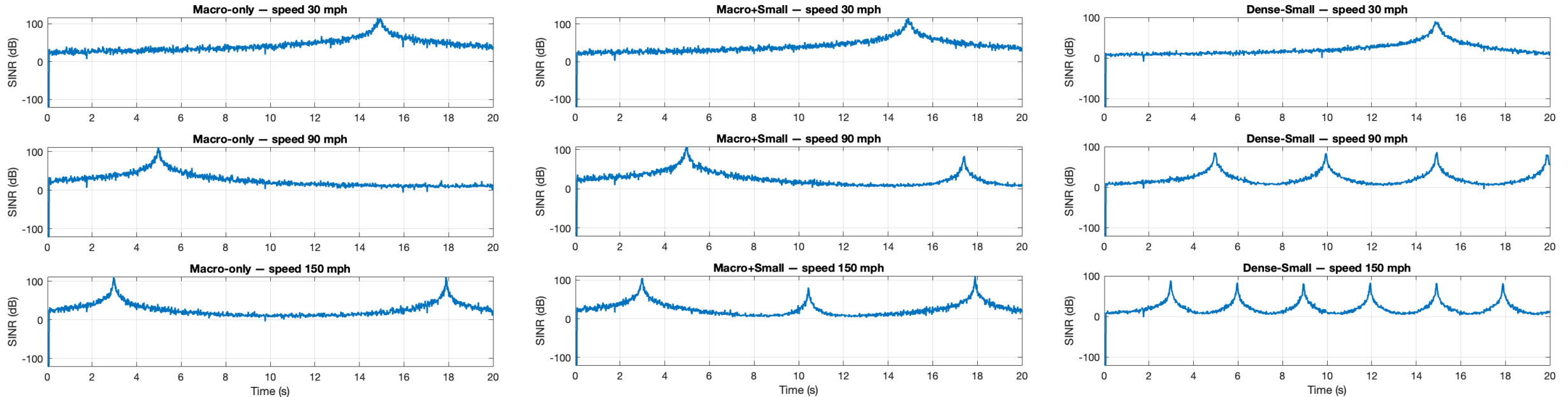
Radio Link Failure (RLF)

Definition: Instances when SINR drops below a threshold for a certain duration, indicating potential loss of connection.

Code Measurement:

```
if SINR_dB(tt) < RLF_sinr_threshold_dB
    outageTimer = outageTimer + dt;
else
    outageTimer = 0;
end
if outageTimer >= RLF_timeout
    rlf_count = rlf_count + 1;
    outageTimer = 0;
end|
```

Results & Analysis



Possible reasons for SINR peaks

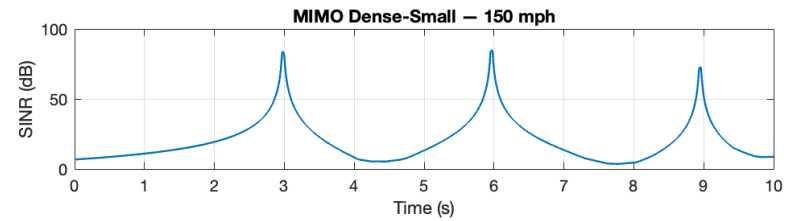
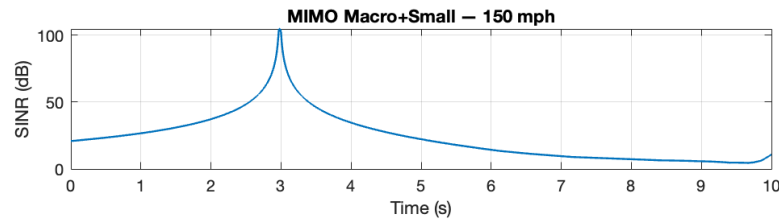
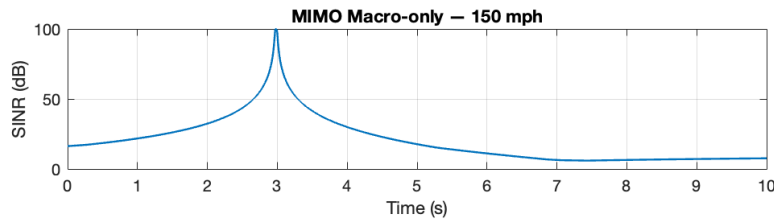
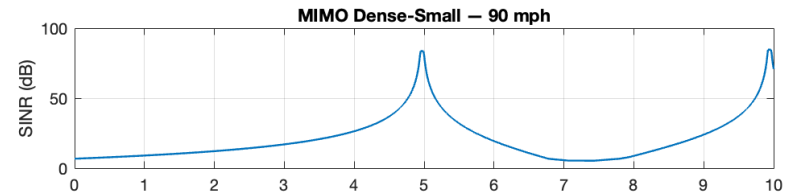
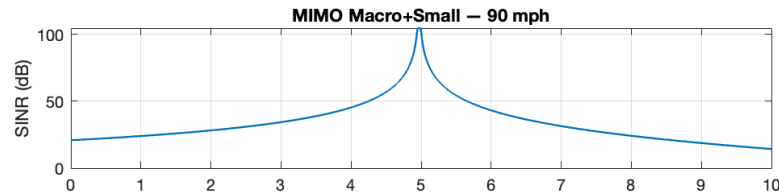
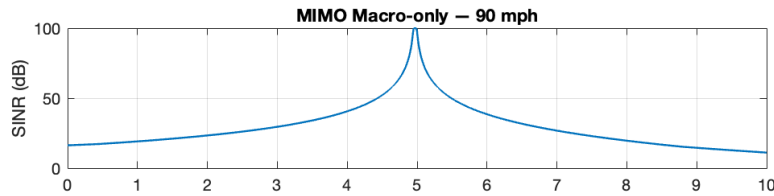
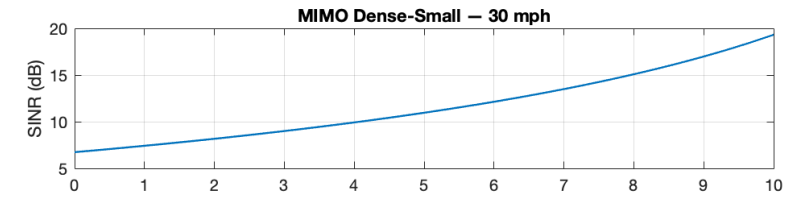
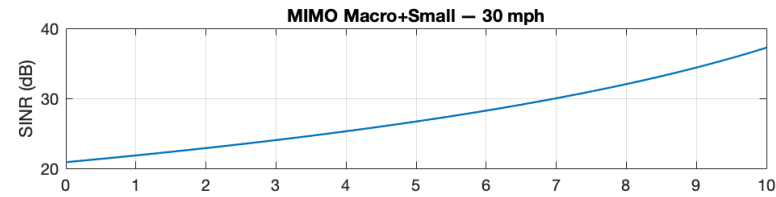
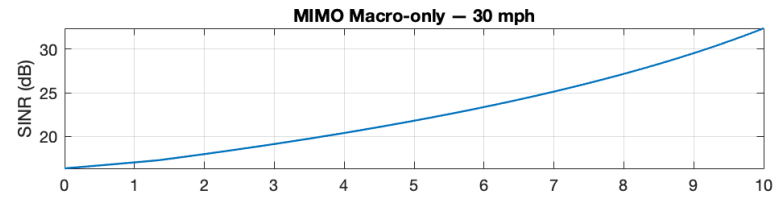
Constructive multipath

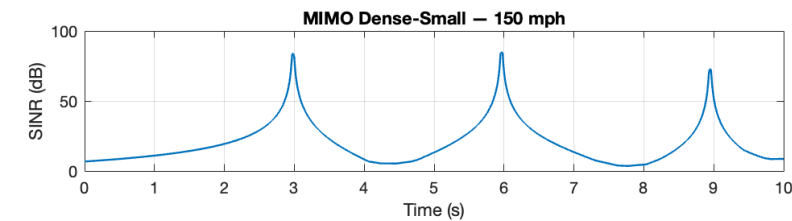
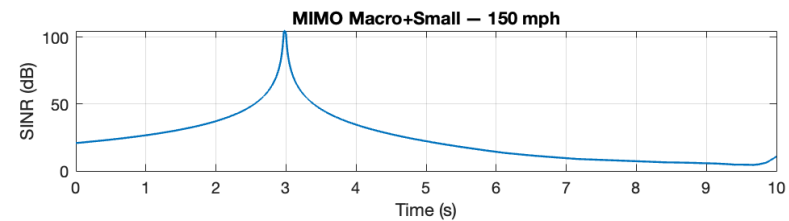
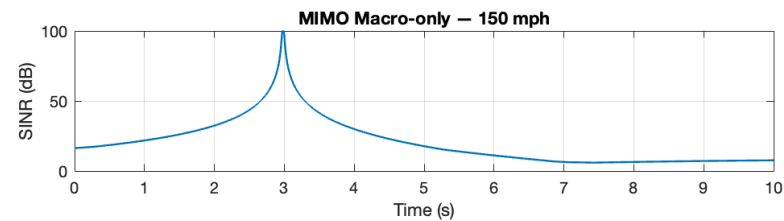
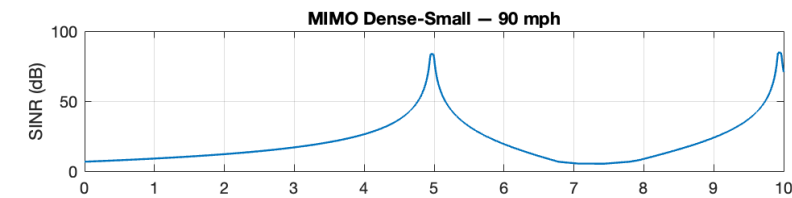
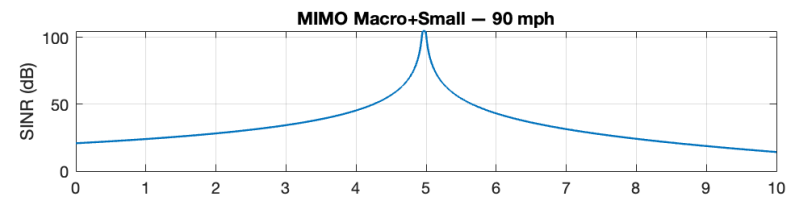
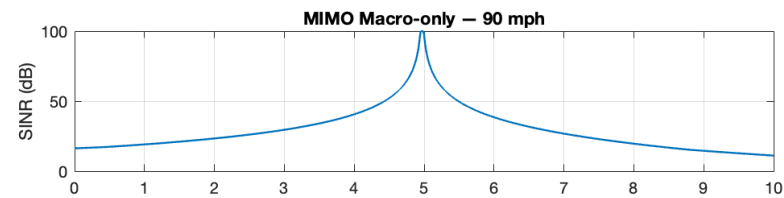
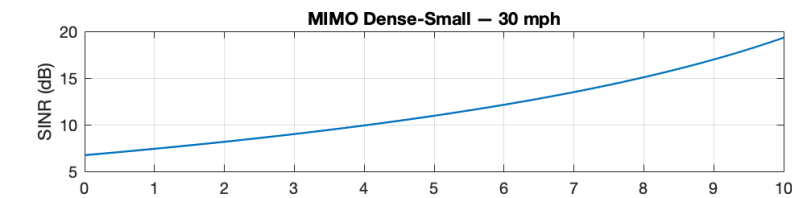
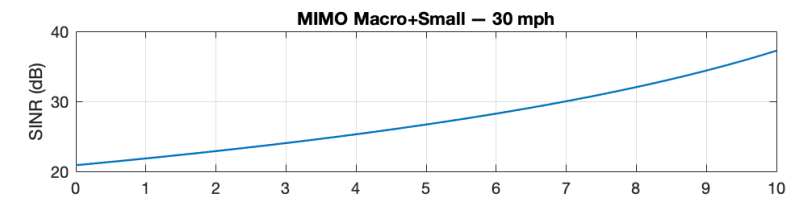
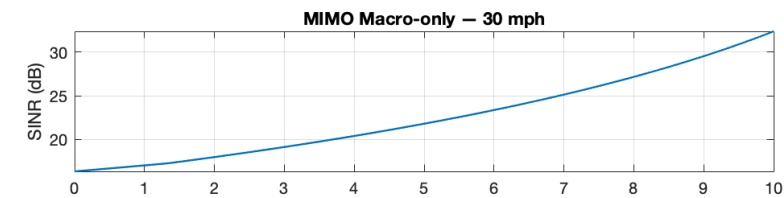
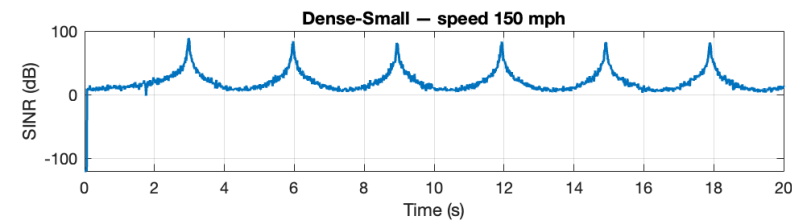
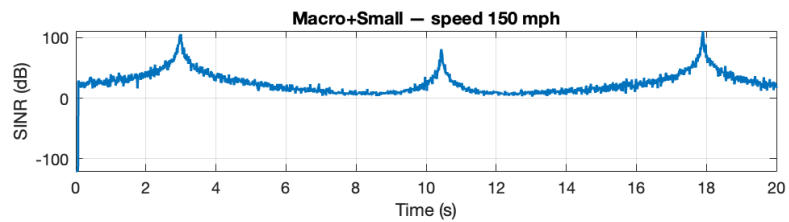
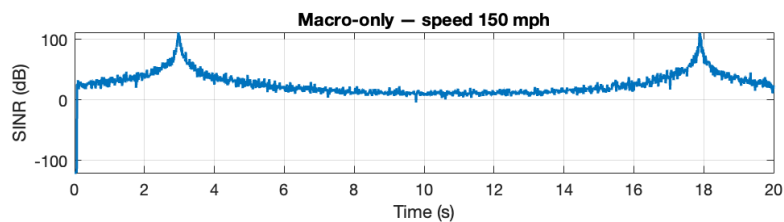
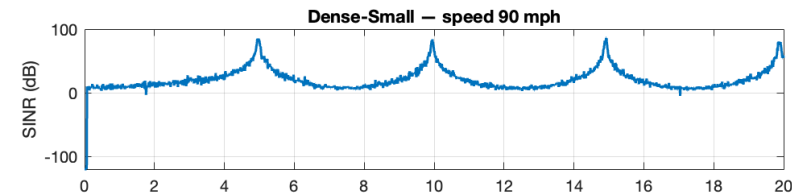
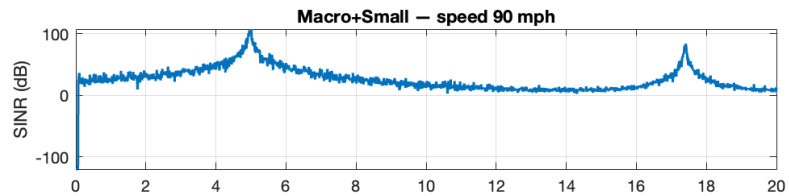
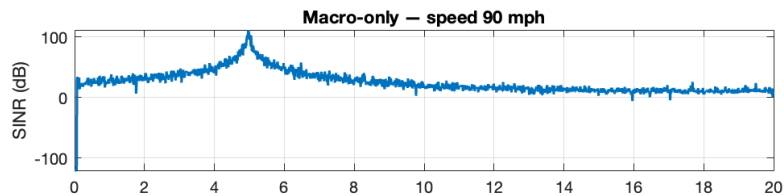
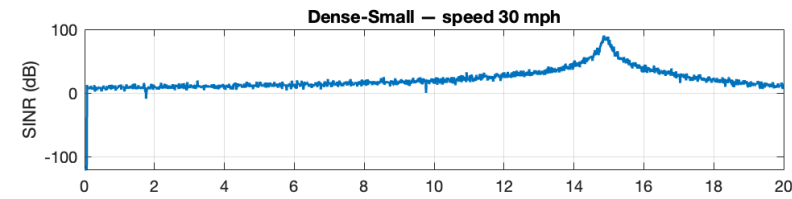
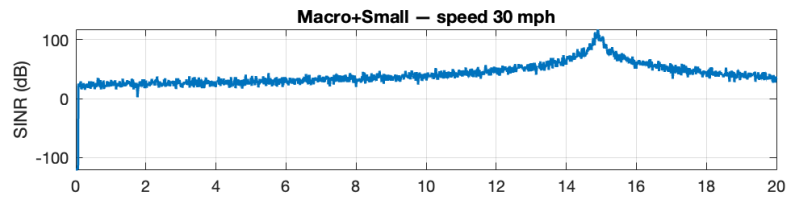
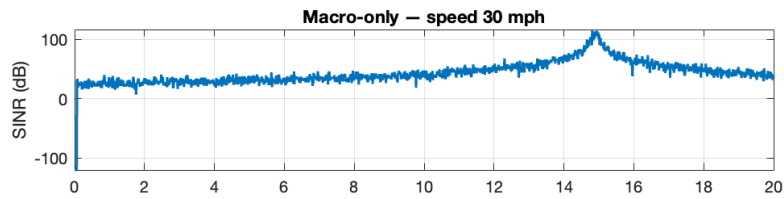
Closer to serving cell

Lower interference — UE moves away from interferers.

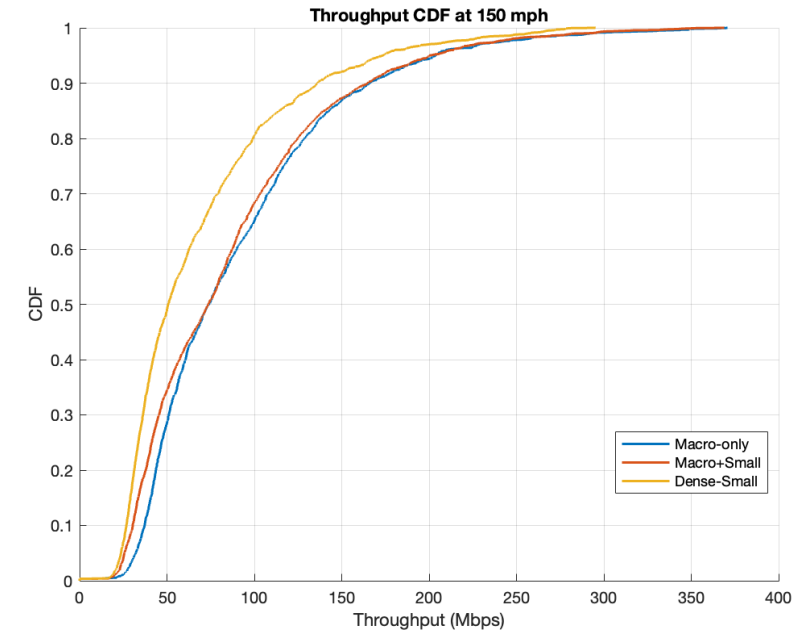
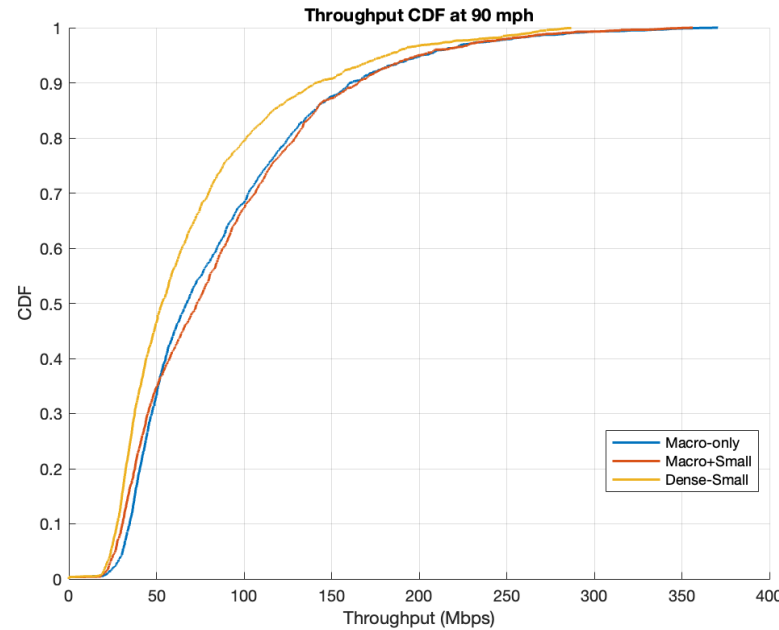
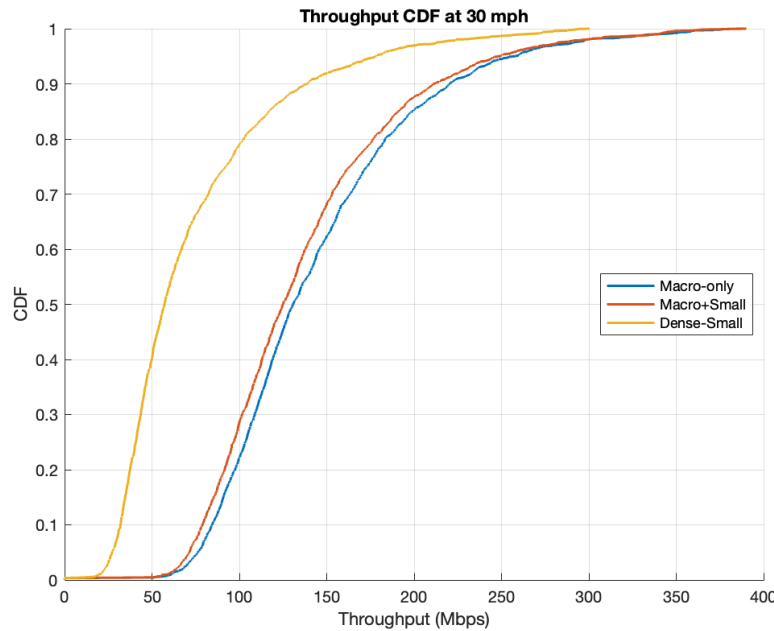
Beam alignment (MIMO) — main lobe hits the UE.

Results & Analysis





Results & Analysis



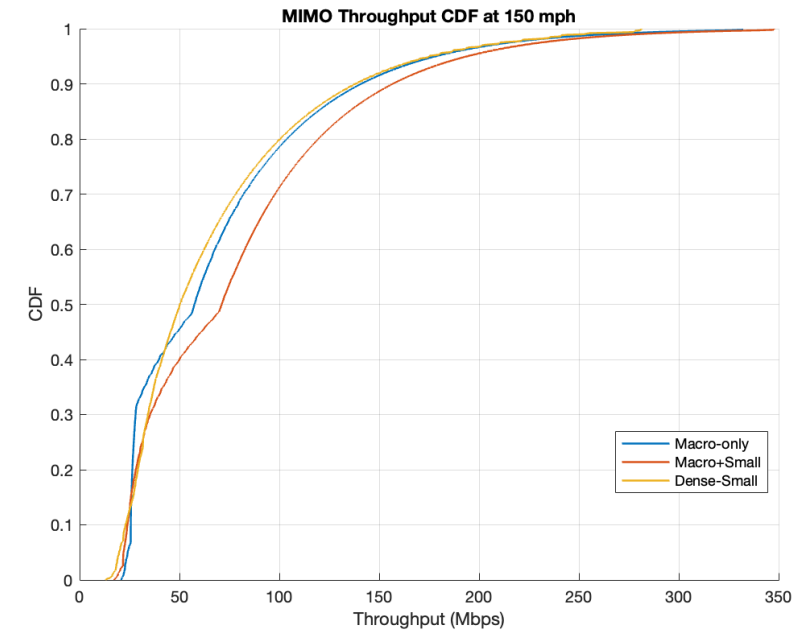
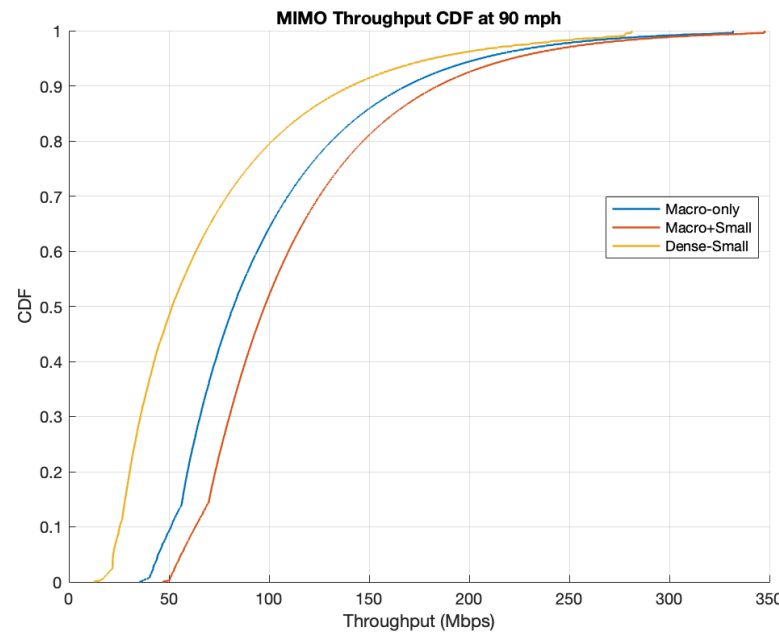
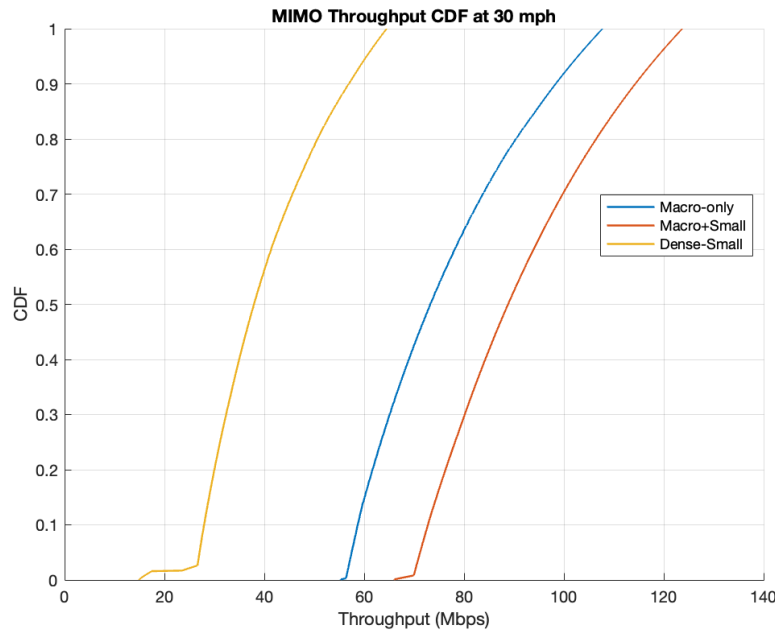
CDF (Cumulative Distribution Function)

Throughput CDF = the percentage of users (or time samples) whose throughput is below a certain value.

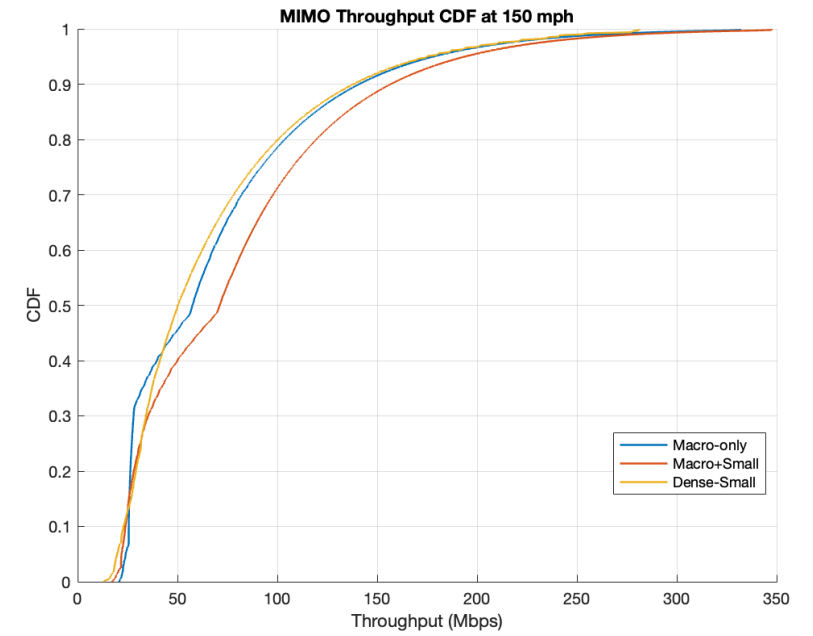
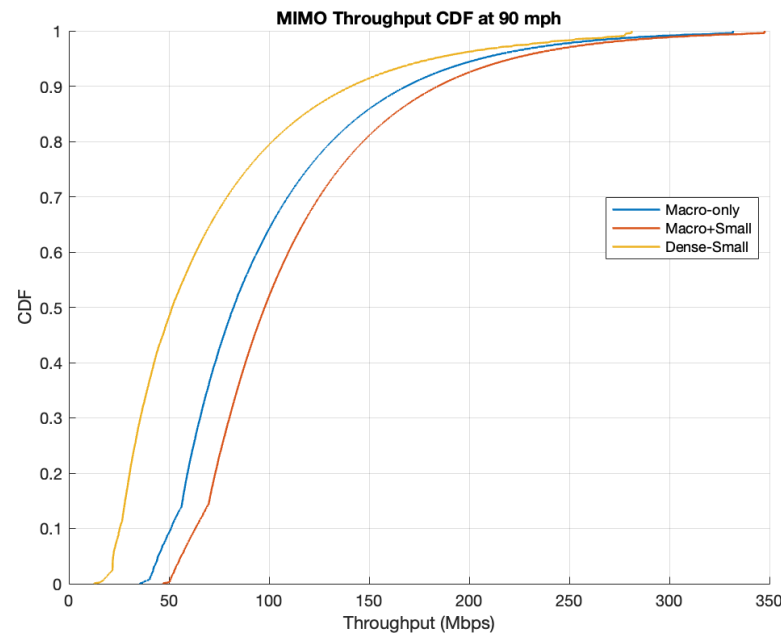
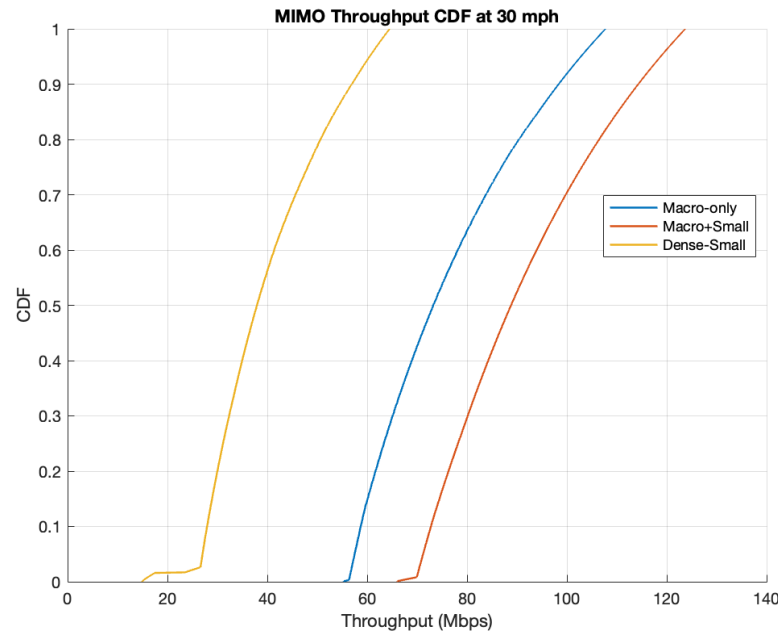
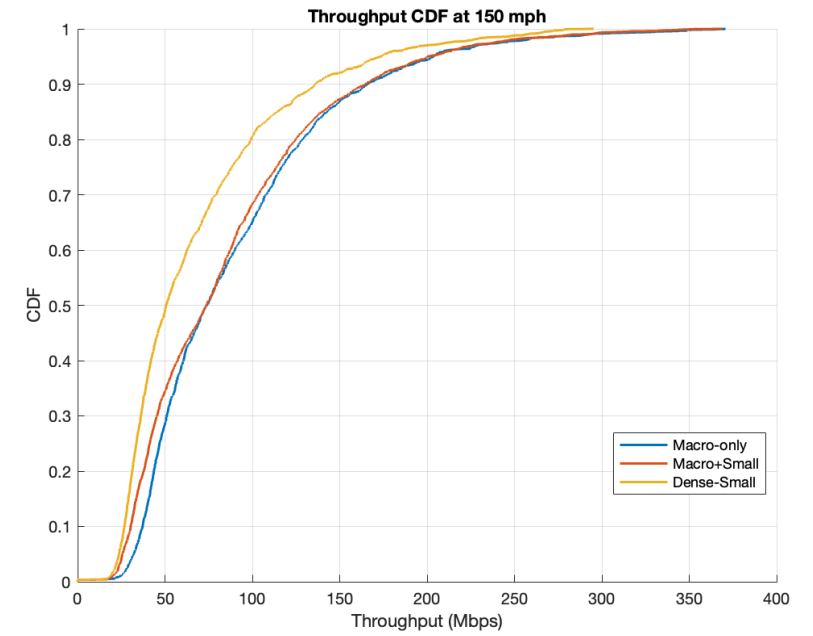
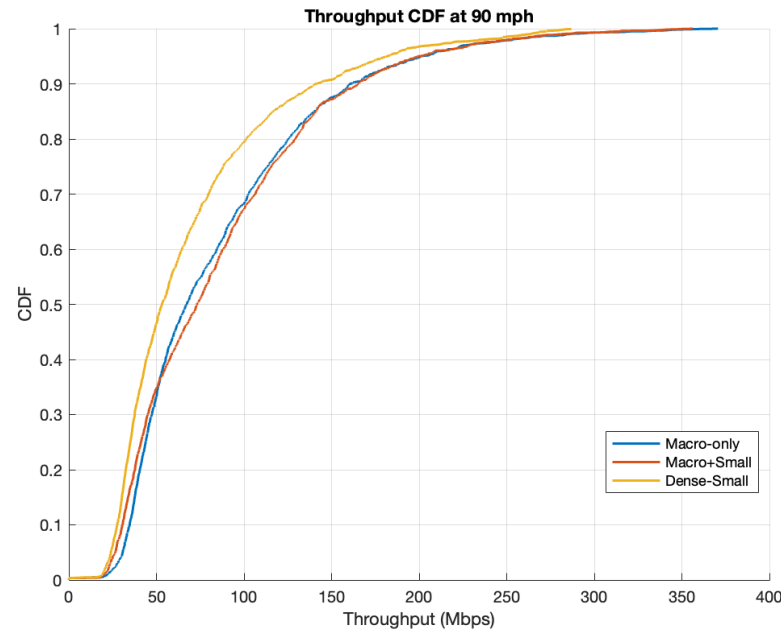
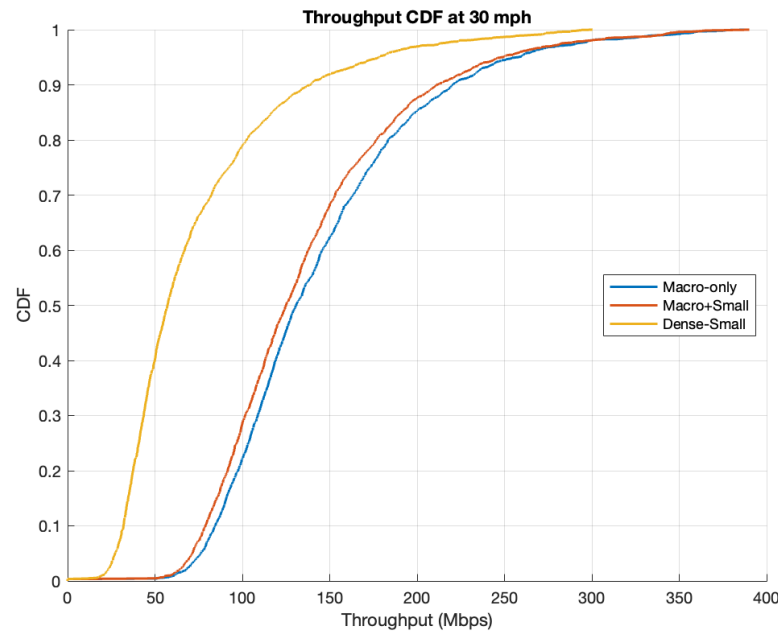
Higher speed → stronger fading → lower SINR

Dense-Small creates heavy interference.

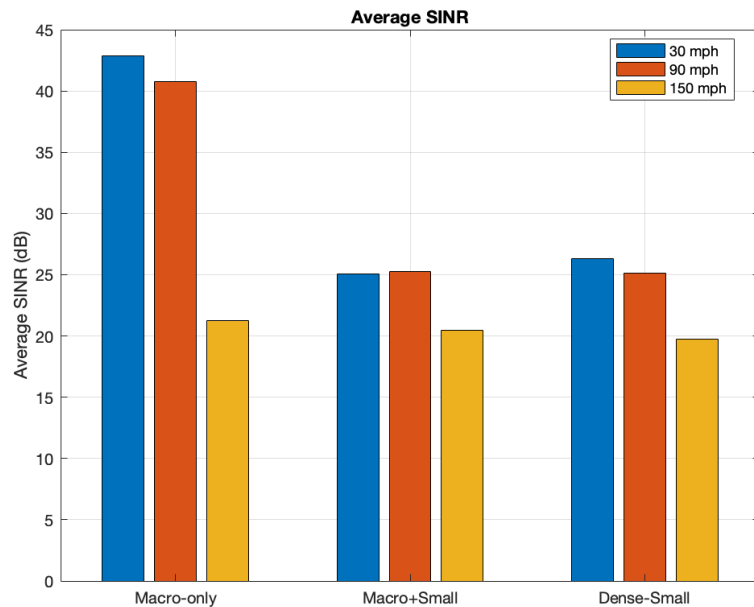
Results & Analysis



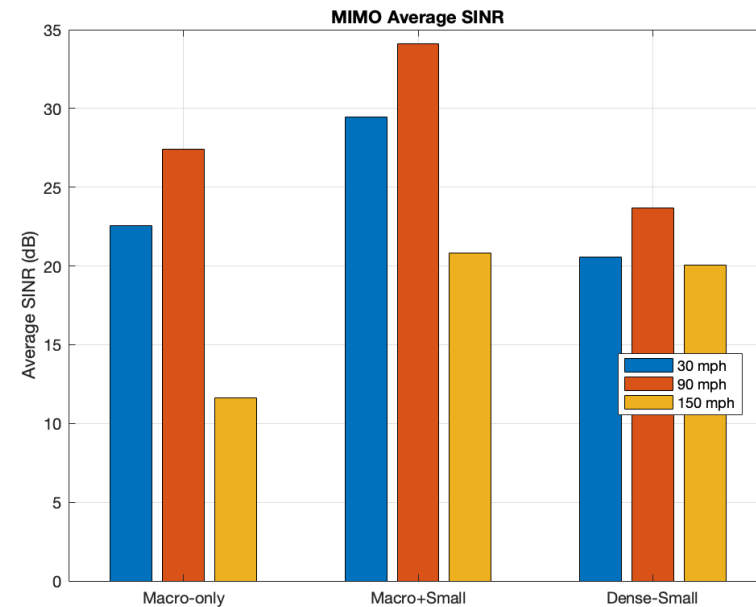
Due to the simplified MIMO model, no beamforming gain at low speed, but more constructive multipath at high speed.



Results & Analysis

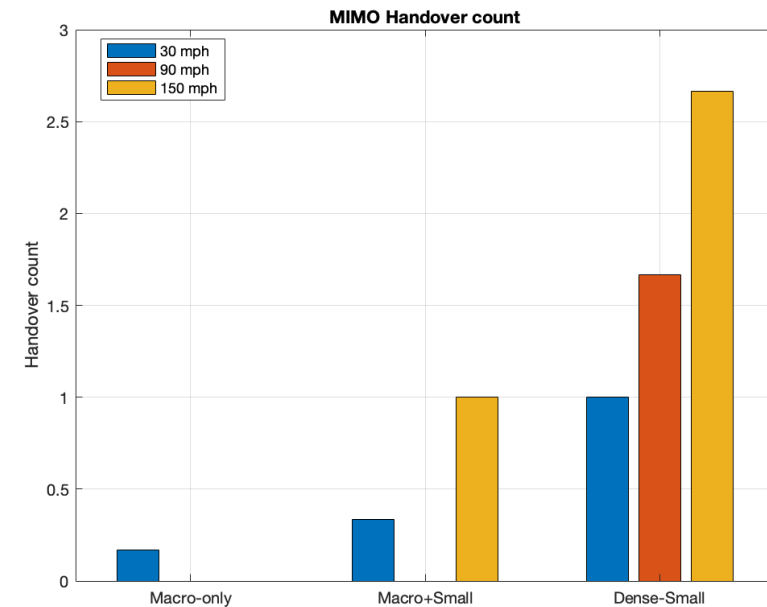
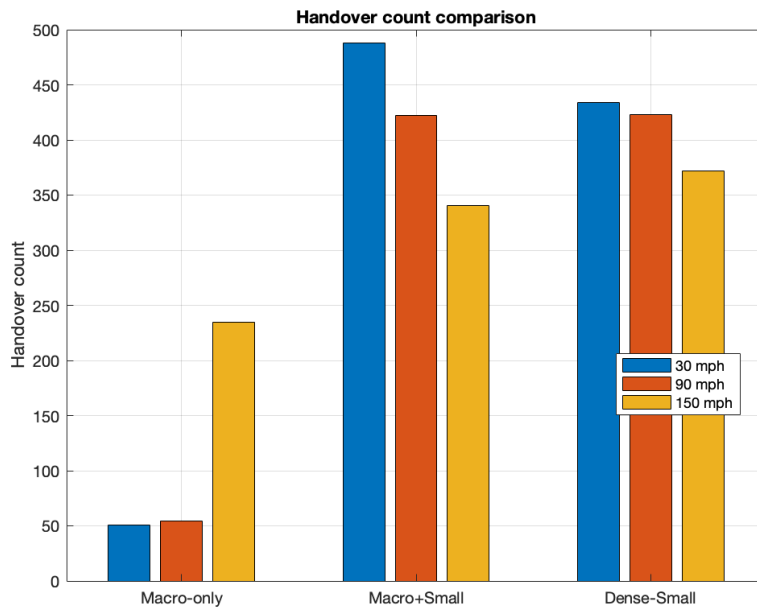


SINR decreases with speed
Fast motion increases fading severity
Macro-only has the least interference.



SINR peaks at 90 mph
A modeling artifact: Doppler +
multipath summation creates a statistical
peak at mid-high speeds.

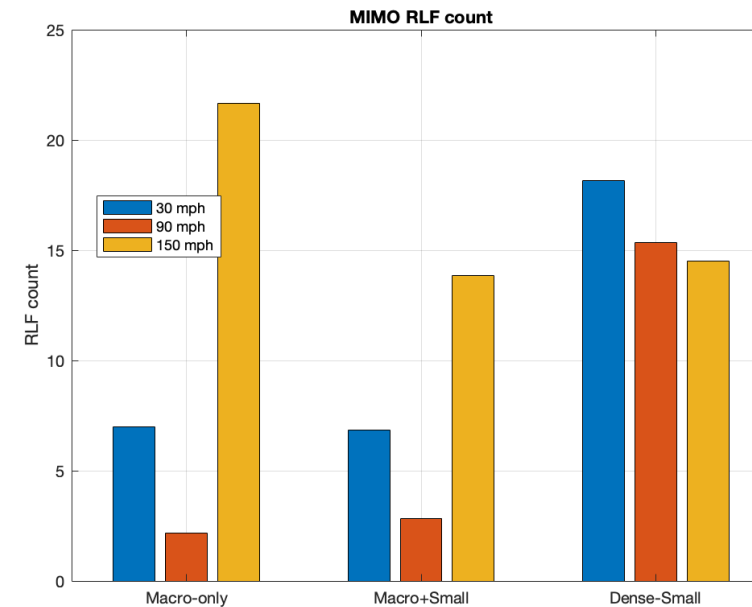
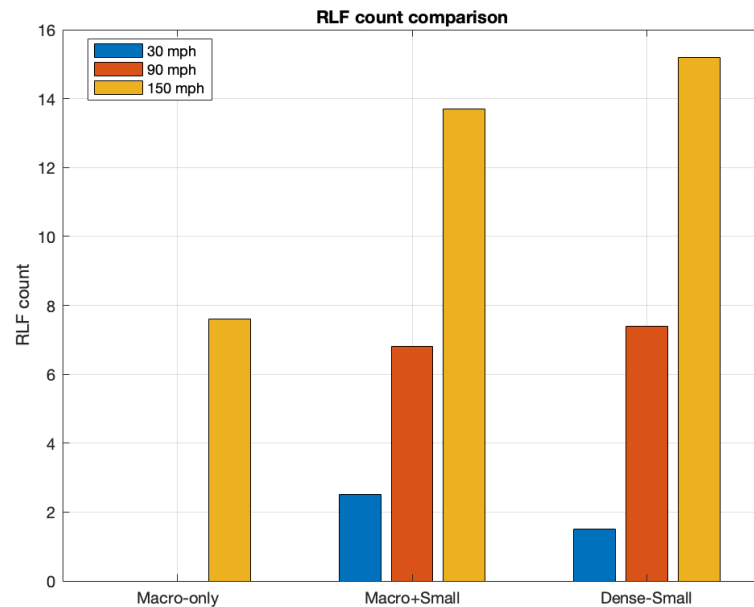
Results & Analysis



MIMO has fewer handovers

MIMO power is smoother → fewer cell crossings; higher speed → more boundary crossings.

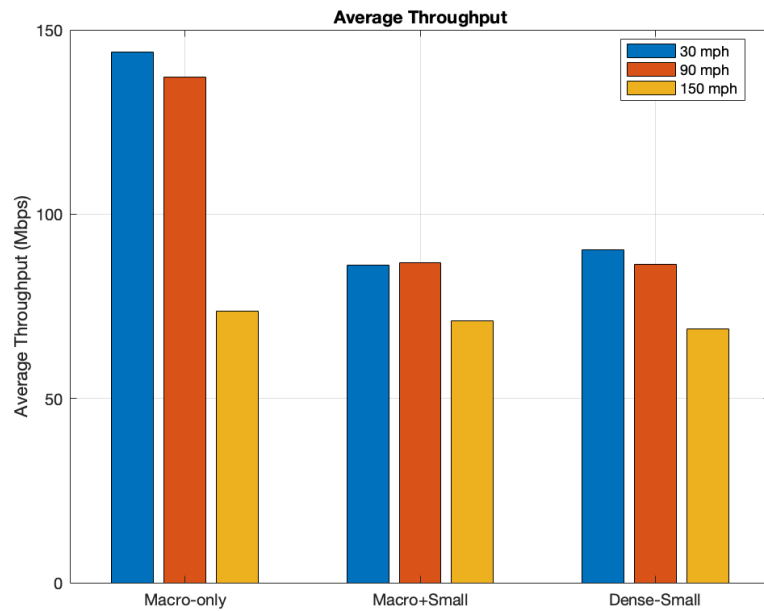
Results & Analysis



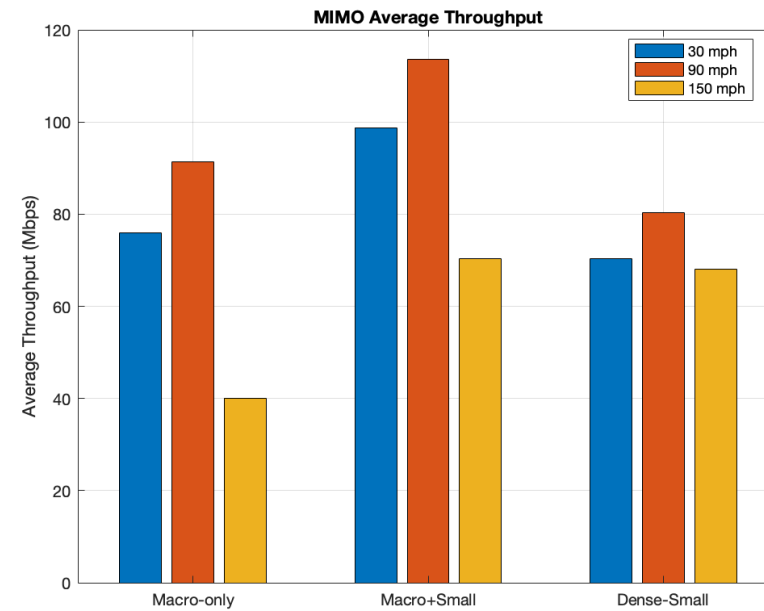
MIMO has more RLFs, especially at high speed

Fewer handovers cause the UE to stay too long on a weak cell → more outages.

Results & Analysis



Macro-only yields the highest throughput
Its wide coverage and low interference floor
lead to more consistent data rates.



MIMO throughput underperforms SISO
Because the model does not include realistic
MIMO gains, while increased RLF events reduce
the averaged rate.

Conclusions

The SINR behavior of SISO and MIMO is similar due to simplified MIMO modeling.

Throughput declines at higher speeds, and macro-only consistently delivers the most reliable rates.

Macro + small cells benefit from additional coverage but suffer more from mobility-driven interference.

MIMO reduces handover frequency but increases RLF events, especially at high speeds.

Mobility, interference level, and network architecture jointly determine end-to-end performance.